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SECTOR AT A GLANCE 09

Industrial Water Management in India

Water in India

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INDUSTRIAL WATER MANAGEMENT IN INDIA

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The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview

Water is vital for India's industrial growth and will play an increasingly crucial role as Indian manufacturing and industrial activity scales up. But industrial activity also drives significant negative externalities for the freshwater ecosystems largely in terms of localized over-extraction and significant chemical and heavy metal pollution. For instance, India's industries meet 80% of their water requirements from underground sources, thus stressing the per capita availability of water.¹ Similarly, with demand from industries like steel, textiles, and power rising rapidly, efficient industrial water management is imperative to balance the ecosystem needs and those of industrial activity.²

India has recognized the importance of managing industrial water effectively. Programs and policies like the National Water Policy, the Environment Protection Act of 1986, and discharge standards of the National Green Tribunal that focus on ensuring sustainable use of water resources within Indian industry. Several state and local-level policies also address sustainable industrial water management.

Going forward, there is a significant opportunity to focus on driving water use efficiency in Indian industry and also overall competitiveness, harnessing wastewater treatment as a business opportunity, creating urban wastewater markets, and improving environmental and human health. To enable this, it will be essential to mitigate challenges poor monitoring and compliance related to industrial water usage and discharge, lack of appropriate financing and technology support for SMEs to move to clean production practices.

Specific areas that would benefit from focused policy attention include real time industrial effluent monitoring in key river systems, innovations for localised zero liquid discharge systems that are especially tailored for Indian SMEs in polluting sectors such as textiles, and investing in capacity building within Indian industry.³

Figure 1: Key industrial water management statistics

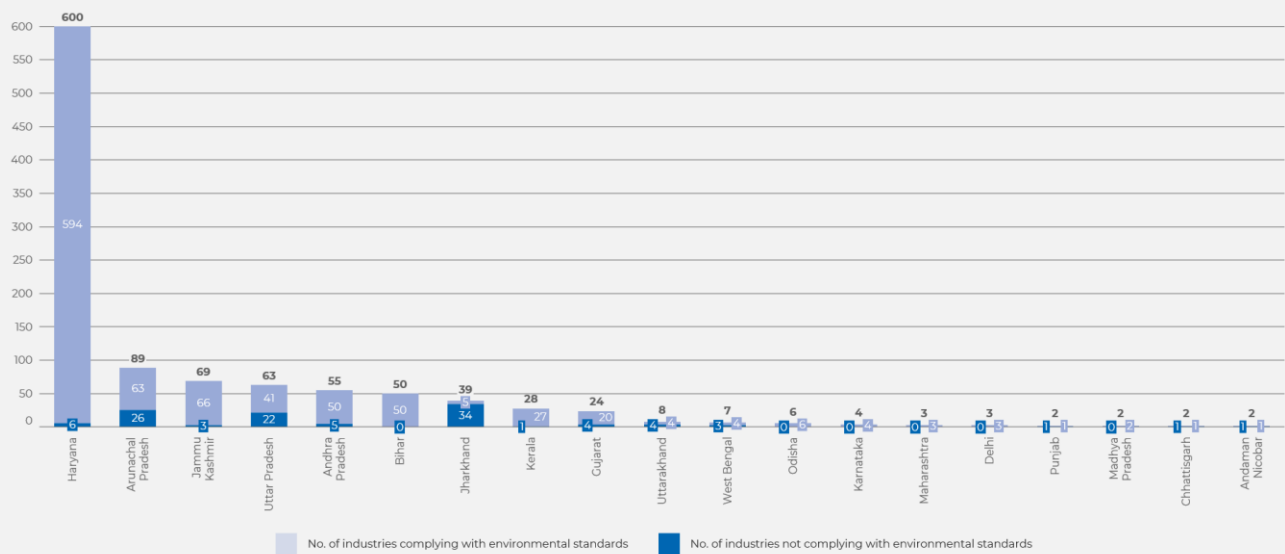


¹ Association of Geospatial Industries, Potential of Geospatial Technologies for the Water Sector in India, 2021

² MoHUA, Industrial Water Supply - Incentives for Conservation, 2017

³ Note: Wastewater is generated from both domestic sewage (i.e., from water supplied for domestic use, which is treated by sewage treatment plants) and industrial use (i.e., from water supplied for industrial purposes and treated by common effluent treatment plants in the case of small industries). See ORF, Arresting India's Water Crisis: The Economic Case for Wastewater Use, 2021.

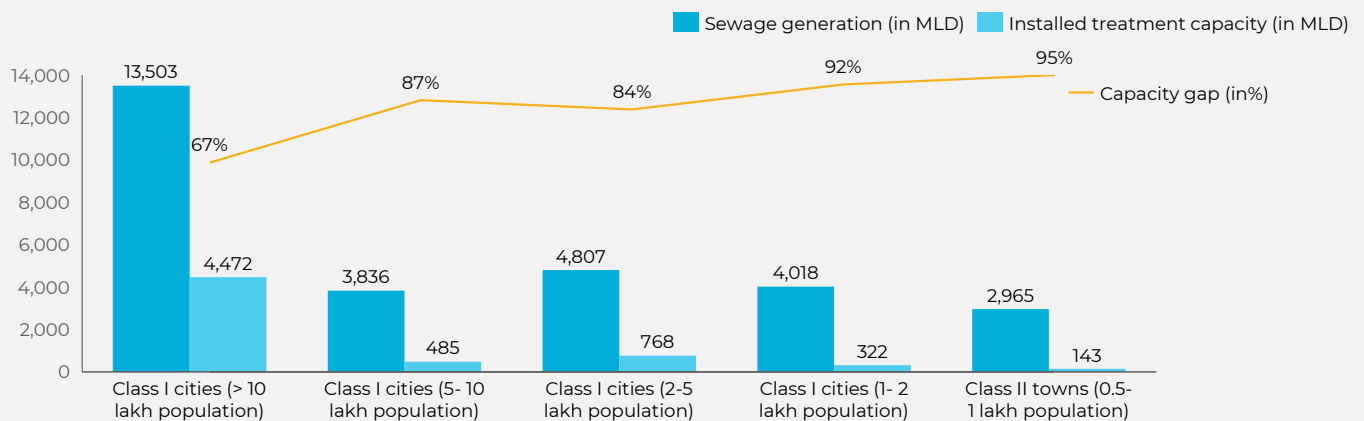
Figure 2: Number of state-wise industries complying with environmental standards out of the number of operational industries (2022)



Note: Assam, Chandigarh, Daman & Diu and Dadra & Nagar Haveli, Goa, Himachal Pradesh, Lakshadweep, Manipur, Mizoram, Sikkim, Tamil Nadu, and Tripura SPCBs/PCs have reported 'zero' operational industries

Source: PIB, Water Pollution, 2022

Figure 3: Wastewater generation and treatment capacity gap in Class I cities and Class II towns (2022)



Source: NITI Aayog, Urban Wastewater Scenario in India, 2022

Opportunities and challenges

Opportunities

This section highlights the outcomes policymakers can achieve by adopting effective industrial water management practices. These revolve primarily around the potential for sustaining water supply, saving costs, protecting the environment, and improving health outcomes.



Treating wastewater to meet industrial water demand:

Regulating use and adopting efficiency measures, reuse, and recycling can reduce stress on freshwater resources, even as industrial demand grows. If the country can utilize 80% of its untreated wastewater from 110 of its most populous cities, 75% of the projected industrial water demand can be met by 2025.⁴ Transitioning to alternative sources like treated wastewater or desalination eases the burden on rivers and aquifers.⁵



Industrial wastewater treatment is a growth opportunity:

With increased manufacturing and process industry growth in India, the demand for treated water is likely to grow at a CAGR of approximately 10% and it could become as large as INR 17,000 crores by 2025.⁶



Improved human and environmental health outcomes through industrial water management:

Industrial activity in India releases a range of untreated heavy metals (lead, chromium, arsenic, zinc), organic compounds (solvents, dyes, petroleum), nutrients (especially from fertilizer factories) into water bodies. The cost of water pollution to public health in India may be over USD 6.7-8.7 billion per year, highlighting the economic benefits of policy solutions that can balance industrial needs and impacts.⁷ Limiting pollution and encouraging sustainable sourcing and usage of water is a significant opportunity to enable healthier environments, communities, and economy in India.⁸



Increase in water use efficiency and improved industrial competitiveness:

There is significant potential for Indian industry to reduce its water footprint through the upgrade of equipment, process changes, water reuse, behaviour changes across the value chain, as well as better use of data and analytics across the supply chain.⁹ Centres of excellence such as the CII-Triveni Water Institute as well as international partners such as IFC that have dedicated programs related to industrial water efficiency can support in this transition. A special focus will be needed to support SMEs in making this transition and packages comprising of technical assistance, financing support, as well as incentives to make this shift will prove invaluable.

Challenges

This section highlights some of the major challenges and constraints facing industrial water management in India. Efforts to resolve such challenges will ensure effective pricing and treatment of water, stricter legislation enforcement and greater internal coordination.

⁴ NITI Aayog, Urban Wastewater Scenario in India, 2022

⁵ NITI Aayog, Urban Wastewater Scenario in India, 2022

⁶ Frost and Sullivan, Indian WWW Treatment Market to Experience a Boom as the Country witnesses increase in Private Investments, 2022

⁷ World Economic Forum, Water pollution is killing millions of Indians. Here's how technology and reliable data can change that, 2019

⁸ World Economic Forum, Water pollution is killing millions of Indians. Here's how technology and reliable data can change that, 2019

⁹ NITI Aayog, Report on Water Neutrality for Indian Industry, 2023



Limited metering and pricing of water:

Water is a state subject, making it challenging to establish fair and uniform industrial water charging policies, as in most states, it is the city-level bodies that look after the water supply.¹⁰ The gradual introduction of volumetric, cost-reflective tariffs can curb excessive use and waste, especially in states like Tamil Nadu, Gujarat, and Maharashtra, that have the highest number of industrial units.¹¹



Insufficient private investment in wastewater management:

Under the Environment (Protection) Act, 1986, and the Water (Prevention & Control of Pollution) Act, 1974, industries and local bodies must set up effluent treatment plants (ETPs)/common effluent treatment plants (CETPs) and sewage treatment plants (STPs) to treat their waste according to environmental standards before releasing it into rivers and water bodies.¹² However, high upfront capital requirements and insufficient awareness of wastewater treatment technologies limit uptake of industrial wastewater management initiatives.¹³ Although, targeted financial assistance programs for small enterprises by states like Madhya Pradesh have shown potential for increasing installation and use of treatment technologies.¹⁴



Limited pollution-control enforcement:

Out of the 351 polluted river stretches, 13% of rivers are severely polluted, with heavy metals from industries threatening the health of both marine and human life.¹⁵ More robust controls and penalties can limit impacts, especially in known hotspots.



Limited institutional coordination:

Limited coordination among institutions across different levels prevents integrated solutions and isolated decision-making stall progress.¹⁶ Platforms engaging stakeholders to share knowledge and collaborate, as piloted in some states, can enable rapid progress through coordinated policy and investments.¹⁷

¹⁰ MoHUA, Review of current practices in determining user charges and incorporation of economic principles of pricing of urban water supply, 2010

¹¹ Ministry of Statistics, Annual Survey of Industries 2019-20, 2020

¹² Ministry of Jal Shakti, Compulsory Water Treatment Plant, 2022

¹³ UN and CEEW, The Perfect Storm - Pathways to Managing India's Water Sector Sustainably, 2018

¹⁴ ORF, The Greening Imperative: How Indian States Are Promoting Sustainable Industry, 2021

¹⁵ CPCB, Polluted River Stretches For Restoration Of Water Quality, 2022, NITI Aayog, Urban Wastewater Scenario in India, 2022

¹⁶ NITI Aayog, Urban Wastewater Scenario in India, 2022

¹⁷ NITI Aayog, Urban Wastewater Scenario in India, 2022

Institutional framework and policy landscape –

Key ministries and departments

This section highlights the major central government bodies and national-level institutions that oversee industrial water management.



Ministry of Jal Shakti

The Ministry aims to effectively manage water as a national resource, which thus includes industrial water use in its ambit. It functions through two departments- the first one being the Department of Water Resources, River Development and Ganga Rejuvenation (DoWR, RD & GR) which is primarily responsible for establishing policy guidelines and programmes for the management, development, and conservation of water. The second department, i.e., the Department of Drinking Water and Sanitation provides technical and financial assistance to states to provide safe and adequate drinking water to rural India.¹⁸

One noteworthy body related to industrial water management is **the Bureau of Water Use Efficiency**, Ministry of Jal Shakti. The Bureau aims to promote efficient use of water in irrigation, domestic water supply, municipal and/or industrial uses in the country through planning and execution of nation-wide programmes. Amongst its various sub-groups lies one for the industries sector as well.¹⁹



Central Pollution Control Board²⁰

The Board is responsible for prevention, control, and abatement of water pollution (thus including pollution caused by industries) in accordance with Water (Prevention and Control of Pollution) Act, 1974.

Major policies and programs

This section highlights existing policies related to industrial water management and their most prominent features. A more comprehensive list of policies is provided in the policy repository of the NITI Knowledge Platform.



National Water Policy:

It aims to enhance water use efficiency in industries by establishing benchmarks, water footprints, and water auditing. It ensures the prevention of industrial effluents and pollutants from contaminating groundwater and encourages desalination where feasible. Additionally, incentives and subsidies are provided to promote industrial pollutant recovery and recycling, supporting sustainable water management.²¹



Environment Protection Act, 1986:

It allows the government to set discharge and emission standards for various pollutants, thereby controlling the quality of water in Indian rivers.²² It also restricts certain areas from specific industries, operations, or processes unless certain safeguards are met.²³



Guidelines for industries to obtain Consent to Establish and Consent to Operate:

As per the Water Act 1974 and the Air Act 1981, individual establishments are required to obtain Consent to Establish before commencement of construction and Consent to

¹⁸ Ministry of Jal Shakti, About the Ministry, accessed in December 2023

¹⁹ Ministry of Jal Shakti, About the Ministry, accessed in December 2023

²⁰ Ministry of Jal Shakti, About the Ministry, accessed in December 2023

²¹ Ministry of Jal Shakti, National Water Policy, 2012

²² Ministry of Environment, Forest and Climate Change, The Environment (Protection) Act, 1986

²³ Ministry of Environment, Forest and Climate Change, The Environment (Protection) Act, 1986

Operate before starting operation from the concerned state pollution control board or committees. Moreover, these boards or committees can ask defaulting units to pay environmental compensation for environmental damage.²⁴



NGT Discharge standards:

The National Green Tribunal has notified discharge standards for industrial sewage treatment plants, including pH levels, Bio-Chemical Oxygen Demand, total suspended solids, etc.²⁵



State-level regulation and partnerships:

Some states, like Maharashtra, have regulatory authorities controlling industrial (and municipal) water supply, treatment, and discharge.²⁶ Moreover, the government has attempted collaboration with industries on R&D, pilot programs, and water infrastructure development to leverage private sector expertise for public benefit.²⁷ Scaling such successful models nationally can drive wider change.

²⁴ CPCB, Mechanisms/ Guidelines for Control of Pollution and Enforcement of Environment Norms, 2020

²⁵ National Green Tribunal, *NS-Deshpande-v-UOI*, 2018

²⁶ NITI Aayog, Urban Wastewater Scenario in India, 2022, MoHUA, Review of current practices in determining user charges and incorporation of economic principles of pricing of urban water supply, 2010

²⁷ NITI Aayog, Urban Wastewater Scenario in India, 2022

Way forward

This section highlights considerations for policymakers seeking to advance policy formulation and implementation in this sector.

Water is a critical raw material for Indian industry and will be significant to driving India's economic and industrial growth going forward. However, care must be taken to avoid significant negative externalities in terms of water pollution, over-extraction, as accompanying harms to human and animal health.

India has to take a carefully balanced approach and focus on key initiatives and programs related to industrial water:

 Strengthening real-time industrial effluent and discharge monitoring	• Strengthening real-time monitoring of industrial effluents and discharges in key river systems • Enforcing penalties linked to non-treated wastewater recharge by industries to deter environmental violations.
 Scaling ZLD systems in the SME sector	• Design and scale context-appropriate zero liquid discharge (ZLD) systems within Indian industries by providing dedicated financing and technical assistance for ZLD implementation in small and medium enterprises (SMEs) in polluting industries (e.g., textiles, leather).
 Fostering market-based solutions	• Encourage the creation of local city-level wastewater markets where treated industrial wastewater gets reused for various purposes (e.g., agriculture). • Leverage wastewater markets, including through regulatory instruments such as the Green Credit Programme that support voluntary corporate action on water conservation, to create valuable incentives for companies to invest in sustainable treatment infrastructure.
 Ensuring accurate measurement and fair pricing for water usage	• Enforce volumetric metering for industrial water usage through technology-enabled monitoring and data analytics to ensure accurate measurement and fair pricing.



**Fostering
industry
collaborations**

- Support and scale activities of institutes like the CII-Triveni Water Institute, promoting partnerships institutes, industries, and industrial clusters.
- Promote "race-to-the-top" dynamics through collaborative ESG benchmarking programs and similar initiatives.

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Recommended readings

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<https://www.trade.gov/market-intelligence/india-water-and-wastewater-treatment-industry>



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